Appendix B

PDE Miscellanea

B.1 Classification of PDE

Partial differential equations may be classified in many different ways. One particular classification that has proven useful, is that of second order PDE into elliptic, parabolic, and hyperbolic equations. This naming scheme is inspired by the classification of conic sections, with which it has some similarities.

B.1.1 Conic Sections

Conic sections are precisely what the name suggests they are: curves that are defined by the intersection of a plane with a cone. This is illustrated in Fig. B.1. Mathematically, the two-

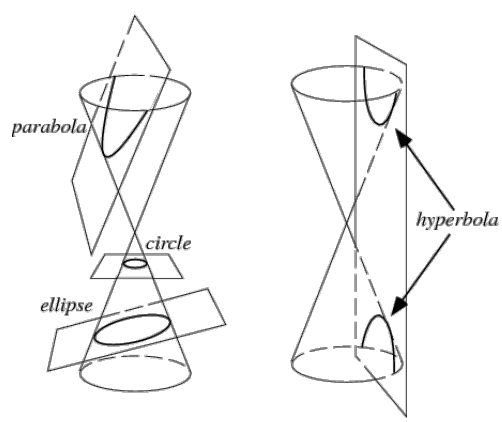

Figure B.1: Conic sections. (Weisstein, Eric W. "Conic Section." From MathWorld–A Wolfram Web Resource. `http://mathworld.wolfram.com/ConicSection.html`)

dimensional curves obtained by these intersections can be represented by the formula

$$ax^2 + bxy + cy^2 + dx + ey + f = 0, \tag{B.1}$$

where different types of conic sections are obtained as follows:

- $ac - 4b^2 > 0$: Ellipse
- $ac - 4b^2 < 0$: Hyperbola
- $ac - 4b^2 = 0$: Parabola

Note that the last of these conditions is only a necessary condition for the conic section (B.1) to be a parabola. (The expression $ac - 4b^2 = 0$ is not sufficient to define a parabola. For example, *all* coefficients could be zero under this definition.) For our purposes it will be enough, however, to note this necessary condition.

Example B.1 Let $a = 1/\alpha^2 > 0$ and $c = 1/\beta^2 > 0$. Furthermore, let $f = -1$, and let all other coefficients in (B.1) be zero. Then we obtain

$$\left(\frac{x}{\alpha}\right)^2 + \left(\frac{y}{\beta}\right)^2 = 1.$$

This is an ellipse with semi axes of length α and β.

Example B.2 Let $a = 1/\alpha^2 > 0$ and $c = -1/\beta^2 < 0$. Furthermore, let $f = -1$, and let all other coefficients in (B.1) be zero. Then we obtain

$$\left(\frac{x}{\alpha}\right)^2 - \left(\frac{y}{\beta}\right)^2 = 1.$$

This is a hyperbola with focal points $c = \pm\sqrt{\alpha^2 + \beta^2}$.

Example B.3 Let $a = 1$ and $e = f = -1$. Furthermore, let all other coefficients in (B.1) be zero. Then we obtain

$$y = 1 + x^2.$$

which defines a parabola.

B.1.2 PDE of second order

Let $x = (x_1, \ldots, x_d)^T$, $d \geq 1$, and consider the linear second order PDE

$$-\sum_{i,j=1}^{d} a_{ij} u_{x_i x_j} + \sum_{i=1}^{d} b_i u_{x_i} + cu = f, \tag{B.2}$$

where we assume for simplicity that the coefficients a_{ij}, b_i, and c are constant. If the function u is smooth enough, we have $u_{x_i x_j} = u_{x_j x_i}$, and hence we can take $A = (a_{ij})_{i,j=1,\ldots d}$ to be a symmetric matrix.[1] We then define the PDE as

- elliptic, if A is definite, i.e. all eigenvalues have the same sign,
- hyperbolic, if one eigenvalue of A has opposite sign, compared to all others,
- parabolic, if A is singular, but the matrix $(A|\mathbf{b}) \in \mathbb{R}^{d\times(d+1)}$ has full rank.

[1] We can simply "split" the coefficients in symmetric form. For example, if $d = 2$ and $(x_1, x_2) = (x, y)$, and we are given a term $3u_{xy}$, we can write $3u_{xy} = 3/2u_{xy} + 3/2u_{yx}$.

In the last statement the matrix $(A|\mathbf{b})$ is obtained by adding the vector $\mathbf{b} = (b_1, \ldots, b_d)^T$ as an additional column to the matrix A. (the "minus" sign in front of the second-order terms is simply a convention. The classification does not depend on it.)

Example B.4 Let $d = 2$, let $(x_1, x_2) \equiv (x, y)$, and write a general second order PDE as

$$-au_{xx} - bu_{xy} - cu_{yy} + \alpha u_x + \beta u_y + \gamma u = f.$$

By comparing coefficients we can write this in the form of (B.2), with $a_{11} = a$, $a_{22} = c$, and $a_{12} = a_{21} = b/2$. Thus, the matrix A is written

$$A = \begin{pmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{pmatrix}.$$

The eigenvalues are determined from the characteristic equation $\det(A - \lambda I) = 0$, which leads to

$$\lambda_\pm = \frac{a+c}{2} \pm \frac{1}{2}\sqrt{(a+c)^2 - (4ac - b^2)}.$$

If $4ac - b^2 = 0$, it is clear that there will be one zero eigenvalue and hence the equation is parabolic, provided that the matrix

$$\begin{pmatrix} a & \frac{b}{2} & \alpha \\ \frac{b}{2} & c & \beta \end{pmatrix}$$

has full rank. If $4ac - b^2 > 0$ (respectively $4ac - b^2 < 0$) we have two positive roots (respectively one positive and one negative root), and the PDE is elliptic (respectively hyperbolic). The similarity to the classification of conic sections is obvious. One should, however, not overemphasize this connection. Rather, one should think of it as merely a naming convention.

Example B.5 Consider the heat equation

$$u_t = au_{xx} \qquad \Leftrightarrow \qquad -au_{xx} + u_t = 0,$$

where $a > 0$. Let $(x_1, x_2) \equiv (x, t)$. Written in the generic form of (B.2), we have $a_{11} = a$, $a_{22} = a_{12} = a_{21} = 0$. Hence the matrix A is given by

$$\begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix},$$

which is clearly singular. Furthermore, we have $b_1 = 0$ and $b_2 = 1$ for the coefficients multiplying the first derivatives. Hence, the matrix $(A|\mathbf{b})$ is given by

$$\begin{pmatrix} a & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Clearly the rank is equal to 2 for any $a > 0$. Hence the heat equation is parabolic.

Example B.6 Consider the wave equation

$$u_{tt} = c^2 u_{xx} \qquad \Leftrightarrow \qquad c^2 u_{xx} - u_{tt} = 0,$$

where c is a coefficient, which we assume constant here. Let $(x_1, x_2) \equiv (x, t)$. Written in the generic form of (B.2), we have $a_{11} = -c^2$, $a_{22} = 1$, and $a_{12} = a_{21} = 0$. Hence the matrix A is given by

$$\begin{pmatrix} -c^2 & 0 \\ 0 & 1 \end{pmatrix}$$

The eigenvalues are given by $\lambda_1 = -c^2 < 0$, and $\lambda_2 = 1 > 0$. Hence the equation is hyperbolic.

The restriction to constant coefficients is mainly for convenience here. Similar ideas apply to the case of non-constant coefficients. However, if the coefficients depend on x, the PDE may *change* type in different parts of the computational domain.

B.2 Dirichlet Problem for the Heat Equation

We give a derivation of the solution to the problem

$$\begin{aligned} \frac{\partial u}{\partial t} &= a\frac{\partial^2 u}{\partial x^2}, & (x,t) \in (0,L) \times (0,\infty) \\ u(x,0) &= \phi(x), & x \in [0,L] \\ u(0,t) &= 0, & t > 0, \\ u(L,t) &= 0, & t > 0. \end{aligned} \tag{B.3}$$

We assume that ϕ is a continuous function that satisfies the boundary conditions (i.e. $\phi(0) = \phi(L) = 0$). We try *separation of variables*. This means we are looking for a solution of the form

$$u(x,t) = T(t)X(x).$$

Upon substitution into the heat equation (B.3) one obtains

$$T'X = aX''T.$$

We can write

$$\frac{T'}{aT} = \frac{X''}{X} = -\lambda = const.$$

Since the left-hand side depends only on t, and the right-hand side depends only on x, we can conclude that both must be constant. (We set the constant as $-\lambda$ for later convenience.) We can thus reduce the problem to the solution of two ODEs. Firstly we have

$$T' = -\lambda a T$$

which has the general solution

$$T(t) = C_1 e^{-\lambda a t}, \tag{B.4}$$

where C_1 is some constant. The second ODE is

$$X'' + \lambda X = 0, \qquad X(0) = X(L) = 0. \tag{B.5}$$

For $\lambda > 0$, this is the well-known differential equation for the harmonic oscillator. The ODE admits the general solution

$$X(x) = D_1 \sin(\sqrt{\lambda}x) + D_2 \cos(\sqrt{\lambda}x)$$

Using the boundary condition $X(0) = 0$ we immediately find $D_2 = 0$. The condition $X(L) = 0$ implies now that we must have

$$\sqrt{\lambda} = \frac{k\pi}{L}, \qquad k = 1, 2, \ldots,$$

and thus

$$X(x) = D_1 \sin\left(\frac{\pi k x}{L}\right) \tag{B.6}$$

solves (B.5). By not allowing $k = 0$ we have excluded the uninteresting trivial solution $X \equiv 0$. Using $\lambda \leq 0$ also leads to the trivial solution: For $\lambda = 0$, we get $X'' = 0$, and hence the solution X is affine linear. The only such function satisfying the boundary conditions is $X \equiv 0$. For $\lambda < 0$ the solution to (B.5) is exponential $X(x) = D_1 e^{-\sqrt{-\lambda}x} + D_2 e^{\sqrt{-\lambda}x}$. The boundary conditions can only be satisfied, if $D_1 = D_2 = 0$.

We thus have infinitely many solutions to the problem that satisfy the boundary conditions at $x = 0$ and $x = 1$. Combining (B.4) and (B.6), the most general solution is thus given by a linear combination

$$u(x,t) = \sum_{k=1}^{\infty} A_k e^{-a\frac{k^2\pi^2}{L^2}t} \sin\left(\frac{k\pi x}{L}\right). \tag{B.7}$$

Since there does not seem an obvious reason to stop the sum at any particular value of k, we have daringly extended the sum all the way to infinity. The mathematically inclined student will immediately realize that this raises additional questions related to the convergence of the infinite sum. We will justify this a posteriori when we use the initial conditions to determine the remaining constants. Formally it is easy to see that we have a valid solution if we can select the coefficients A_k such that

$$\phi(x) = u(x,0) = \sum_{k=1}^{\infty} A_k \sin\left(\frac{k\pi x}{L}\right). \tag{B.8}$$

Such a series representation is called a Fourier sine series. Again proceeding formally we note that upon integrating both side of (B.8) against $\sin\left(\frac{\pi k x}{L}\right)$ and noting that

$$\int_0^L \sin\left(\frac{\pi k x}{L}\right) \sin\left(\frac{\pi l x}{L}\right) dx = \begin{cases} 0 & l \neq k \\ \frac{L}{2} & l = m, \end{cases}$$

we obtain that *if* the sum converges the coefficients must be given by

$$A_k = \frac{2}{L} \int_0^L \phi(x) \sin\left(\frac{k\pi x}{L}\right) dx.$$

The issue of convergence of such sums is a well-studied topic in Fourier analysis. A review of the theory is beyond our scope. Suffice it to say that whether or not the representation we have derived here is proper depends on the smoothness of the function ϕ. It is known from Fourier analysis and the theory of parabolic differential equations that the assumption on $\phi(x)$ that we have made here (i.e. that ϕ be continuous and satisfy the boundary conditions) are enough. The series representation converges uniformly, and we can exchange summation and differentiation in (B.7), which makes it trivial to check that indeed our solution satisfies the heat equation. (Each term in (B.7) does!) In fact, one can show that the solution is infinitely many times differentiable. This is true even if the initial conditions are only continuous.

B.3 The Wave Equation

We consider the wave equation as a pure initial value problem (a so-called *Cauchy Problem*):

$$u_{tt} = c^2 u_{xx}, \qquad x \in \mathbb{R},\ t > 0, \tag{B.9}$$

$$u(x,0) = \phi(x), \qquad x \in \mathbb{R}, \tag{B.10}$$
$$u_t(x,0) = \psi(x), \qquad x \in \mathbb{R}. \tag{B.11}$$

Note that any function $u(x,t) = f(x-ct) + g(x+ct)$, where f and g are twice differentiable, solves (B.9):

$$\begin{aligned} u_{tt} &= c^2 \left(f''(x-ct) + g''(x+ct)\right) \\ c^2 u_{xx} &= c^2 \left(f''(x-ct) + g''(x+ct)\right), \end{aligned}$$

where we have simply used the chain rule of differentiation. It can be shown that, in fact, all solutions of the wave equation must be of this form. Hence, we try to find the solution to the Cauchy problem (B.9) by setting for the initial conditions at $t = 0$:

$$u(x,0) = f(x) + g(x) = \phi(x), \tag{B.12}$$
$$u_t(x,0) = c\left(-f'(x) + g'(x)\right) = \psi(x). \tag{B.13}$$

Differentiate (B.12) to obtain

$$f'(x) + g'(x) = \phi'(x) \qquad \Longleftrightarrow \qquad f'(x) = \phi'(x) - g'(x). \tag{B.14}$$

Substituting this into eq. (B.13) we have

$$\psi(x) = c\left(2g'(x) - \phi'(x)\right) \qquad \Longleftrightarrow \qquad g'(x) = \frac{1}{2}\left(\frac{\psi(x)}{c} + \phi'(x)\right).$$

Using this result in (B.14) we obtain for f'

$$f'(x) = \frac{1}{2}\left(\phi'(x) - \frac{\psi(x)}{c}\right).$$

Integration, say between 0 and x, leads to

$$\begin{aligned} f(x) &= \frac{1}{2}\phi(x) - \frac{1}{2c}\int_0^x \psi(\xi)d\xi + C \\ g(x) &= \frac{1}{2}\phi(x) + \frac{1}{2c}\int_0^x \psi(\xi)d\xi + D, \end{aligned}$$

where we have collected all constants of integration in C and D. Finally we have

$$u(x,t) = f(x-ct) + g(x-ct) = \frac{1}{2}\left(\phi(x-ct) + \phi(x+ct)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(\xi)\, d\xi.$$

(We must have $C + D = 0$ if we are to satisfy the initial conditions (B.12) and (B.13).)

Note that the wave equation has finite speed of propagation. The solution at any location (x,t) is only affected by the initial conditions in the interval $[x-ct, x+ct]$. Conversely, the initial conditions at any x_0 can only influence a region bounded by the straight lines $x_0 \pm ct = const$. These lines are called *characteristics*.

Also note, that the regularity of the solution at (x,t) is now directly determined by the regularity of the initial conditions. For example, u cannot have more derivatives than ϕ.